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/*
With Scripter, you can use JavaScript to create your own custom MIDI
processing
effects, including slider controls for real-time interaction.

For detailed information about using Scripter, including code samples,
see the MIDI plug-ins chapter of the Logic Pro X Effects or
MainStage 3 Effects manual.
*/
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```
// example: simple pass through and MIDI monitor
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```
function HandleMIDI(event)
{
    event.trace();
    event.send();
}
```

```
// Logic Scripter: Raag MIDI Filter
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// Copy-paste this into a Scripter plugin on your MIDI track
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// -----
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// 1. Raag definitions
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// Each Raag is an array of allowed MIDI note offsets relative to C
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// (0 = C, 1 = C#/Db, ..., 11 = B)
```

```
var raags = {
    "Yaman":      [0,2,4,6,7,9,11],
    "Bhairav":    [0,1,4,5,7,8,11],
    "Bhairavi":   [0,1,3,5,7,8,10],
    "Kafi":       [0,2,3,5,7,9,10],
    "Todi":       [0,1,3,6,7,8,11],
    "Marwa":      [0,1,3,5,6,8,10],
    "Bageshree":  [0,1,3,5,6,8,10],
    "Jaunpuri":   [0,1,3,5,6,8,10],
    "Desh":       [0,2,4,5,7,9,11],
    "Hamsadhwani": [0,2,4,5,7,9,11],
    "Bairagi":    [0,1,3,5,7,9,11],
    "Bihag":      [0,2,4,5,7,9,10],
    "Bihari":     [0,2,4,5,7,9,11],
    "Tilang":     [0,2,4,5,7,9,11],
    "Khamaj":     [0,2,4,5,7,9,10],
    "Purvi":      [0,1,3,5,6,8,11],
    "Multani":    [0,1,3,5,6,8,11],
    "Shree":      [0,1,4,5,7,8,11],
    "Shankara":   [0,2,4,6,7,9,11],
    "Madhyamavati": [0,2,4,5,7,9,10],
    "Kalyani":    [0,2,4,6,7,9,11],
    "Ahir Bhairav": [0,1,4,5,7,8,11],
    "Bageshri Kanada": [0,1,3,5,6,8,10],
    "Puriya Dhanashree": [0,1,3,6,7,8,11],
    "Chandrakauns": [0,3,5,8,10],
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    "Darbari Kanada": [0, 1, 3, 5, 7, 8, 10],
    "Malkauns": [0, 3, 5, 8, 10],
    "Nat Bhairav": [0, 1, 4, 5, 7, 8, 11],
    "Komal Rishabh Asavari": [0, 2, 3, 5, 7, 9, 10]
};

// 🏯 East Asian Modes Library
var eastAsianModes = {

    // --- CHINESE MODES ---
    "gong": [0, 2, 4, 7, 9],
    "shang": [0, 2, 5, 7, 10],
    "jue": [0, 3, 5, 8, 10],
    "zhi": [0, 2, 5, 7, 9],
    "yu": [0, 3, 5, 7, 10],
    "bian_gong": [0, 1, 4, 7, 9],
    "bian_zhi": [0, 2, 5, 7, 8],

    // --- JAPANESE MODES ---
    "yo": [0, 2, 4, 7, 9],
    "in": [0, 1, 5, 7, 8],
    "in_sen": [0, 1, 5, 7, 10],
    "hirajoshi": [0, 2, 3, 7, 8],
    "iwato": [0, 1, 5, 6, 10],
    "kumoi": [0, 2, 3, 7, 9],
    "hon_kumoi": [0, 1, 5, 7, 8],

    // --- KOREAN MODES ---
    "pyeongjo": [0, 2, 4, 7, 9],
    "gyemyeonjo": [0, 3, 5, 7, 10],
    "ujo": [0, 2, 4, 7, 9],
    "gyemyeon": [0, 3, 5, 7, 10],

    // --- HYBRID / REGIONAL ---
    "ryukyu": [0, 4, 5, 7, 9],
    "mongolian": [0, 2, 4, 7, 9],
    "anhemitonic": [0, 2, 4, 7, 9],
    "slendro_like": [0, 2, 4, 7, 9],
    "pelog_like": [0, 1, 3, 7, 8]
};

// 🎵 WEST ASIAN MODES (Maqam / Dastgah / Makam)
var westAsianModes = {

    // --- ARABIC MAQAMAT ---
    "maqam_rast": [0, 2, 4, 5, 7, 9, 11],
    "maqam_bayati": [0, 1, 3, 5, 7, 8, 10],
    "maqam_hijaz": [0, 1, 4, 5, 7, 8, 11],
    "maqam_nahawand": [0, 2, 3, 5, 7, 8, 10],
    "maqam_kurd": [0, 2, 3, 5, 7, 9, 10],
    "maqam_saba": [0, 1, 3, 4, 7, 8, 10],

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"maqam_ajam":      [0, 2, 4, 5, 7, 9, 11],
"maqam_huzam":     [0, 1, 3, 5, 7, 8, 11],
"maqam_sikah":     [0, 1, 3, 5, 7, 8, 10],
"maqam_hijazkar":  [0, 1, 4, 5, 7, 8, 11],

// --- PERSIAN DASTGAH ---
"dastgah_shur":    [0, 1, 4, 5, 7, 8, 10],
"dastgah_homayun": [0, 2, 3, 5, 7, 8, 11],
"dastgah_mahur":   [0, 2, 4, 5, 7, 9, 11],
"dastgah_segah":   [0, 1, 4, 5, 7, 8, 11],

// --- TURKISH MAKAM ---
"makam_rast":      [0, 2, 4, 5, 7, 9, 11],
"makam_nihavent":  [0, 2, 3, 5, 7, 9, 11],
"makam_hicaz":     [0, 1, 4, 5, 7, 8, 11],
"makam_ussak":     [0, 1, 3, 5, 7, 8, 10],
"makam_kurdili_hicazkar": [0, 1, 4, 5, 7, 8, 11]
};
var allScales = Object.assign({}, raags, eastAsianModes,
westAsianModes);

// -----
// 2. Parameters
var ParameterNames = ["Raag"];
var ParameterValues = [Object.keys(allScales)];
var activeRaagName = "Yaman"; // default: first Raag
var activeRaagNotes = allScales[activeRaagName];

// -----
// 3. MIDI Event Handler
function HandleMIDI(event) {
    if (event instanceof NoteOn || event instanceof NoteOff) {
        var note = snapToRaag(event.pitch);
        event.pitch = note;
        event.send();
    } else {
        event.send();
    }
}

// -----
// 4. Snap function
function snapToRaag(midiNote) {
    var octave = Math.floor(midiNote / 12);
    var noteInOctave = midiNote % 12;

    // Find closest note in Raag
    var closest = activeRaagNotes[0];
    var minDiff = 12;
    for (var i=0; i<activeRaagNotes.length; i++) {

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        var diff = Math.abs(noteInOctave - activeRaagNotes[i]);
        if (diff < minDiff) {
            minDiff = diff;
            closest = activeRaagNotes[i];
        }
    }
    return octave*12 + closest;
}

// -----
// 5. Parameter Changed Handler
function ParameterChanged(param, value) {
    if (param == 0) { // Raag dropdown
        activeRaagName = ParameterValues[0][value];
        activeRaagNotes = allScales[activeRaagName];
    }
}

```